Digestive System: the GI map

Build this as we go. Every stop on the tube in order, and every organ that pours into it. Competencies 109 to 125.

Name _____

THE TUBE: FOLLOW THE BITE

1. Mouth (oral cavity)
Epithelium: stratified squamous
Happens here: chewing, saliva, bolus
Boundaries: lips, cheeks, palate, tongue, fauces

↓

2. Pharynx
Food uses: oropharynx, laryngopharynx
Sealed off: nasopharynx (uvula lifts)
Gate out: upper esophageal sphincter (skeletal)

↓

3. Esophagus
Epithelium: stratified squamous
Outer layer: adventitia
Through the diaphragm at: T10, esophageal hiatus
Gate out: lower esophageal (cardiac) sphincter

↓

4. Stomach
Regions in order: cardia, fundus, body, pyloric part
Epithelium: simple columnar; gastric pits and glands
Extra muscle layer: oblique
Folds: rugae
Gate out: pyloric sphincter

↓

5. Small intestine
Duodenum, 25 cm: C around the pancreas; bile and juice enter at the major papilla
Jejunum, 1 m: most absorption
Ileum, 2 m: Peyer patches
Surface: circular folds, villi, microvilli
Gate out: ileocecal valve

↓

6. Large intestine
In order: cecum, ascending, transverse, descending, sigmoid
Only here: teniae coli, haustra, omental appendices
Epithelium: simple columnar, goblet cells, no villi
Job: absorb water, form feces

↓

7. Rectum and anal canal
Rectum: in the pelvis, in front of the sacrum; stores feces
Anal canal lining near the anus: stratified squamous
Two gates: internal (smooth) and external (skeletal) anal sphincters

ACCESSORY ORGANS: WHAT POURS INTO THE TUBE, AND WHERE

Salivary glands, into the mouth
Parotid duct opens: cheek, upper second molar
Submandibular duct opens: beside the lingual frenulum
Sublingual ducts open: floor under the tongue

Teeth and tongue
Tongue muscles: extrinsic move it, intrinsic shape it
Papillae without taste buds: filiform
Adult teeth: 32; 8 incisors, 4 canines, 8 premolars, 12 molars

Liver, into the duodenum by bile
Lobes: right, left, quadrate, caudate
Lobule: hepatocytes, sinusoids, canaliculi, central vein, portal triads
Bile path: canaliculi, hepatic ducts, common hepatic duct, common bile duct, ampulla, duodenum

Gallbladder
Regions: fundus, body, neck
Its duct: cystic duct, two-way
Job: stores and concentrates bile

Pancreas, into the duodenum
Regions: head, body, tail; retroperitoneal
Main duct joins the common bile duct at the ampulla
Accessory duct opens at the minor papilla
Acini (exocrine) and islets (endocrine)

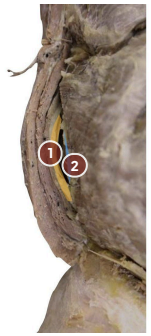
DRAW: THE WHOLE CANAL IN THE TORSO, DIAPHRAGM AT THE TOP, ACCESSORY ORGANS BESIDE IT, A RING AT EVERY SPHINCTER

Digestive System, Part I

Competencies 109 to 114, 120, 121: peritoneum and folds, retroperitoneal organs, the gut wall and its sphincters, mouth, salivary glands, tongue, esophagus.

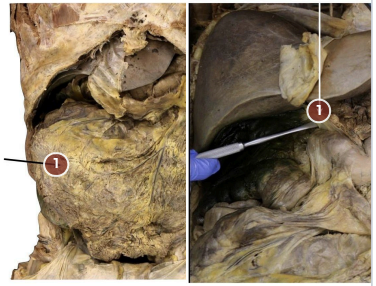
Name _____

A. Peritoneum and the omenta (cadaver) COMPETENCY 111



Cadaver, peritoneum: parietal and visceral layers

1. Parietal peritoneum
2. Visceral peritoneum



Cadaver, greater omentum Cadaver, lesser or

1. Greater omentum
1. Lesser omentum

FIVE FOLDS: WHAT EACH ONE CONNECTS

Greater omentum: greater curvature to the transverse colon

Lesser omentum: lesser curvature to the liver

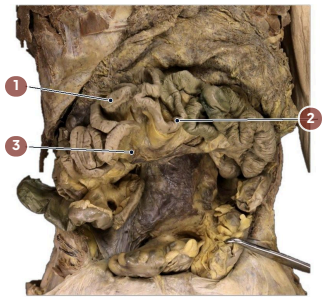
Falciform ligament: liver to the anterior wall and diaphragm

Mesentery: jejunum and ileum to the posterior wall

Mesocolon: transverse and sigmoid colon to the posterior wall

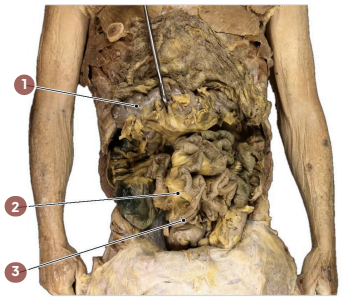
Extra notes and drawings

B. Mesentery, and what sits behind the peritoneum (cadaver) COMPETENCY 111, 112



Cadaver, mesentery of the small intestine

1. Ileum
2. Jejunum
3. Mesentery



Cadaver, transverse colon lifted

1. Transverse colon
2. Mesentery
3. Small intestine

FILL IN

Intraperitoneal: wrapped, hanging on a fold

Retroperitoneal: behind the peritoneum, fixed

Retroperitoneal organs: kidneys, pancreas, duodenum, ascending and descending colon

On a mesocolon: transverse and sigmoid colon

Extra notes and drawings

Digestive System, Part I

Competencies 109 to 114, 120, 121: peritoneum and folds, retroperitoneal organs, the gut wall and its sphincters, mouth, salivary glands, tongue, esophagus.

Name _____

C. The wall of the tube, and its gates (draw it) COMPETENCY 109, 110, 114

FOUR LAYERS, LUMEN OUTWARD

1. mucosa: epithelium, lamina propria, muscularis mucosae
2. submucosa: connective tissue, vessels, submucosal plexus
3. muscularis: inner circular, outer longitudinal, myenteric plexus between
4. serosa (visceral peritoneum), or adventitia on the esophagus

SIX SPHINCTERS, IN THE ORDER FOOD MEETS THEM

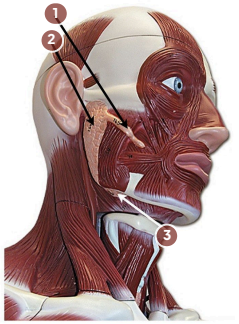
1. upper esophageal (skeletal)
2. lower esophageal, cardiac (smooth)
3. pyloric (smooth)
4. ileocecal (smooth)
5. internal anal (smooth, involuntary)
6. external anal (skeletal, voluntary)

FILL IN

Submucosal plexus controls	secretion
Myenteric plexus controls	movement (motility)
Stratified squamous lines the	mouth, pharynx, esophagus, anal canal
Simple columnar lines the	stomach and intestines

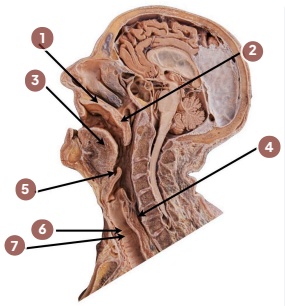
DRAW: A CROSS SECTION OF THE TUBE AS FOUR RINGS, WITH THE TWO PLEXUSES DOTTED IN

D. Mouth, salivary glands, tongue, esophagus COMPETENCY 113, 120, 121



Model, salivary glands

1. Parotid duct
2. Parotid gland
3. Submandibular gland



Cadaver, midsagittal: palate, tongue, esophagus

1. Hard palate
2. Soft palate
3. Tongue
4. Esophagus
5. Epiglottis
6. Vestibular folds (false vocal folds)
7. Vocal folds (true vocal cords)

FILL IN

Parotid duct:	cheek, by the upper second molar
Submandibular duct:	beside the lingual frenulum
Sublingual ducts:	floor, under the tongue
No taste buds:	filiform papillae
Extrinsic muscles:	move the tongue
Intrinsic muscles:	change its shape
Esophageal hiatus:	T10

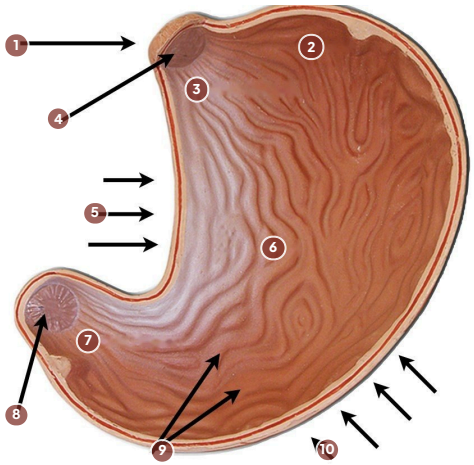
Extra notes and drawings

Extra notes and drawings

Digestive System, Part II

Competencies 115 and 122 to 125: stomach, liver, gallbladder and bile path, pancreas, and the accessory organ disorders. Name _____

A. Stomach regions and features (model) COMPETENCY 115



Model, stomach, interior

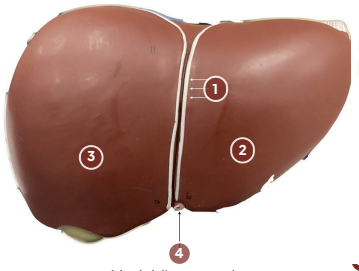
1. Esophagus
2. Fundus
3. Cardiac region
4. Cardiac sphincter
5. Lesser curvature
6. Body of stomach
7. Pyloric region
8. Pyloric sphincter (pylorus)
9. Gastric rugae
10. Greater curvature

FILL IN

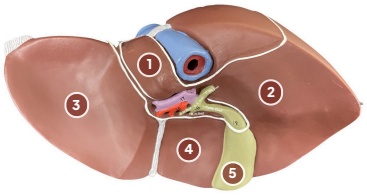
- Muscle layers, inside out: oblique, circular, longitudinal
- Extra layer: oblique
- Lesser curvature faces the liver
- Greater curvature holds the greater omentum
- Rugae are folds of mucosa and submucosa

Extra notes and drawings

B. Liver lobes and ligaments (model) COMPETENCY 122



Model, liver, anterior



Model, liver, inferior

1. Falciform ligament
2. Left lobe of liver
3. Right lobe of liver
4. Ligamentum teres (round ligament)
5. Caudate lobe
6. Right lobe of liver
7. Left lobe of liver
8. Quadrate lobe
9. Gallbladder

FILL IN

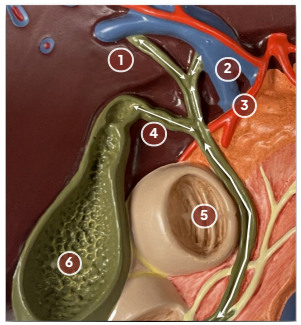
- From the front: right and left lobes
- From below: quadrate and caudate
- Falciform ligament reaches the anterior wall, diaphragm
- Ligamentum teres was the umbilical vein
- Porta hepatis, in: portal vein, hepatic artery
- Porta hepatis, out: hepatic ducts

Extra notes and drawings

Digestive System, Part II

Competencies 115 and 122 to 125: stomach, liver, gallbladder and bile path, pancreas, and the accessory organ disorders. Name _____

C. Bile path and the pancreas (model and cadaver) COMPETENCY 123, 124



Model, gallbladder and the biliary tree

1. Right hepatic duct
2. Left hepatic duct
3. Common hepatic duct
4. Cystic duct
5. Common bile duct
6. Gallbladder

A DROP OF BILE, IN ORDER

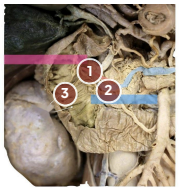
1. canaliculi
2. right and left hepatic ducts
3. common hepatic duct
4. cystic duct, in and out of the gallbladder
5. common bile duct
6. ampulla, sphincter of Oddi
7. major duodenal papilla, duodenum

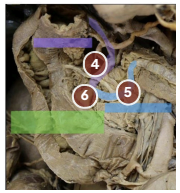
FILL IN

Joins the common bile duct: main pancreatic duct

Exocrine: acini, 99 percent

Endocrine: islets, 1 percent

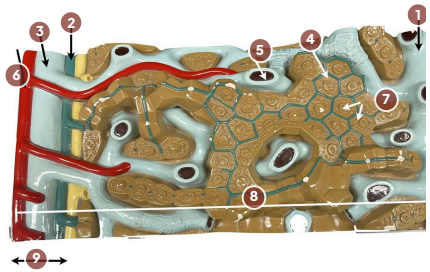




Cadaver, pancreatic and bile ducts entering the duodenum

1. Acc… pan…
2. Main pancreatic duct
3. Plicae circulares
4. Common bile duct
5. Main pancreatic duct
6. Hepatopancreatic ampulla

D. Liver lobule, and the organs in place (model and cadaver) COMPETENCY 122, 125



Model, liver lobule

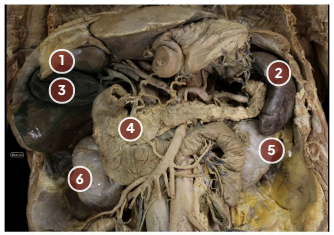
1. Central vein
2. Bile duct
3. Branch of hepatic portal vein
4. Bile canaliculi
5. Liver sinusoids
6. Branch of hepatic artery proper
7. Hepatocytes
8. Liver lobule
9. Portal triad

FILL IN

Blood flows toward the central vein

Bile flows toward the portal triad

Portal triad: portal vein, hepatic artery, bile duct



…daver, major organs of the abdomen

1. Right lobe of liver
2. Spleen
3. Gallbladder
4. Pancreas
5. Left kidney
6. Right kidney

DISORDERS: NAME THE ORGAN

Gallstones, cholecystitis: gallbladder

Hepatitis, cirrhosis: liver

Pancreatitis: pancreas

Mumps: parotid gland

Stone at the ampulla blocks bile and pancreatic juice

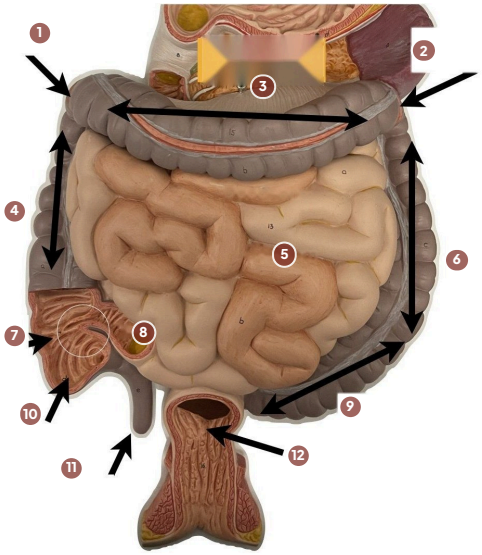
Extra notes and drawings

Extra notes and drawings

Digestive System, Part III

Competencies 116 to 119: small intestine, large intestine and rectum, the path of food, and the canal disorders. Name _____

A. Small and large intestine in place (model) COMPETENCY 116, 117



Model, small and large intestine

1. Hepatic flexure (right colic flexure)
2. Splenic flexure (left colic flexure)
3. Transverse colon
4. Ascending colon
5. Jejunum
6. Descending colon
7. Ileocecal valve
8. Ileum
9. Sigmoid colon
10. Cecum
11. Vermiform appendix
12. Rectum

FILL IN

Duodenum: about 25 cm (10 inches), mostly retroperitoneal

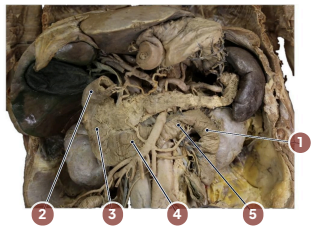
Jejunum: about 1 m (3 feet)

Ileum, the longest: about 2 m (6 feet)

Food leaves the ileum into the cecum, through the ileocecal valve

Four parts of the colon: ascending, transverse, descending, sigmoid

B. Duodenum (cadaver) and the colon surface (model) COMPETENCY 116, 117



Cadaver, the four parts of the duodenum

1. Jejunum
2. Superior part of the duodenum
3. Descending part of the duodenum
4. Inferior (horizontal) part of the duodenum
5. Ascending part of the duodenum

FILL IN

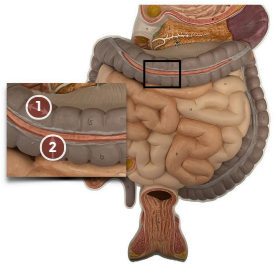
Folding, big to small: circular folds, villi, microvilli

Lacteal absorbs most fats, into lymph

Large intestine only: teniae coli, haustra, omental appendices

Teniae coli are the outer longitudinal layer, three bands

No haustra on the rectum



Model, large intestine surface

1. Haustra
2. Teniae coli

Extra notes and drawings

Extra notes and drawings

Digestive System, Part III

Competencies 116 to 119: small intestine, large intestine and rectum, the path of food, and the canal disorders. Name _____

C. Trace a bite of food, mouth to anus COMPETENCY 118

IN ORDER, WITH A RING AT EACH SPHINCTER

1. oral cavity, fauces
2. oropharynx, laryngopharynx
3. upper esophageal sphincter, esophagus, lower esophageal sphincter
4. cardia, body, pyloric antrum, canal, pylorus, pyloric sphincter
5. duodenum, jejunum, ileum, ileocecal valve
6. cecum, ascending colon, hepatic flexure
7. transverse colon, splenic flexure, descending colon
8. sigmoid colon, rectum, anal canal, internal and external anal sphincters, anus

FILL IN

Not on the path: nasopharynx, fundus, appendix

Most absorption happens in the small intestine

Water is absorbed and feces form in the large intestine

Extra notes and drawings

D. Disorders: which structure does each one affect? COMPETENCY 119

DISORDER: STRUCTURE, WHAT CHANGES

GERD: lower esophageal sphincter; acid comes up, burns behind the sternum

Hiatal hernia: stomach pushes up through the esophageal hiatus

Peptic ulcer: mucosa of stomach or duodenum; erodes outward

Appendicitis: appendix; pain lower right; burst means peritonitis

Diverticulosis: colon wall, usually sigmoid; pouches

IBD: intestinal wall; Crohn any part, ulcerative colitis colon and rectum

Hemorrhoids: veins of the anal canal and lower rectum

Colorectal cancer: mucosa of colon or rectum, from a polyp

DRAW: THE FOUR WALL LAYERS, WITH AN ULCER ERODING FROM THE MUCOSA OUTWARD

Extra notes and drawings

Digestive System: Histology

The tube and the accessory organs under the microscope. Competencies 109, 113, 115 to 117, 121, 122, 124.

Name _____

A. Esophagus and stomach on the slide COMPETENCY 109, 113, 115



Slide, esophagus (right) beside the trachea

Slide photo to add

Slide: stomach wall, gastric pits and glands

FILL IN

Esophagus epithelium: stratified squamous, nonkeratinized

Esophagus outer layer: adventitia, no serosa

Stomach surface epithelium: simple columnar, mucous cells

Gastric pits lead to gastric glands

Chief cells make pepsinogen

Parietal cells make HCl and intrinsic factor

Stomach muscularis: three layers

Extra notes and drawings

B. Small intestine and large intestine on the slide COMPETENCY 116, 117

Slide photo to add

Slide: small intestine, villi and crypts

Slide photo to add

Slide: colon, crypts with goblet cells, no villi

FILL IN

Villi are fingers of mucosa; capillaries and a lacteal inside

Microvilli form the brush border

Crypts (intestinal glands) hold Paneth cells and enteroendocrine cells

Duodenum only, in the submucosa: Brunner glands, alkaline mucus

Ileum only: Peyer patches

Colon mucosa: simple columnar, many goblet cells, no villi

Colon outer muscle: three bands, the teniae coli

Extra notes and drawings

Digestive System: Histology

The tube and the accessory organs under the microscope. Competencies 109, 113, 115 to 117, 121, 122, 124.

Name _____

C. Liver and pancreas on the slide COMPETENCY 122, 124

Slide photo to add

Slide: liver lobule, central vein and portal triad

Slide photo to add

Slide: pancreas, acini with a pale islet

FILL IN

Six-sided unit: lobule

In the center: central vein

At the corners: portal triads

Between the plates: sinusoids, with Kupffer cells

Blood flows in toward the central vein

Bile flows out to the triads

Pancreas: dark clusters are acini (exocrine)

Pale round patches are islets (endocrine)

Extra notes and drawings

D. Salivary gland on the slide, and the four-layer rule COMPETENCY 109, 121

Slide photo to add

Slide: salivary gland, serous and mucous cells

FILL IN

Serous cells make watery saliva with amylase

Mucous cells make mucus

Parotid is all serous

Sublingual is mostly mucous

FOUR LAYERS, ANY SLIDE OF THE TUBE

1. mucosa: epithelium, lamina propria, muscularis mucosae
2. submucosa, with the submucosal plexus
3. muscularis: inner circular, outer longitudinal, myenteric plexus between
4. serosa or adventitia

DRAW: A WEDGE OF GUT WALL AS IT LOOKS ON A SLIDE, LUMEN AT THE BOTTOM, THE TWO PLEXUSES DOTTED IN

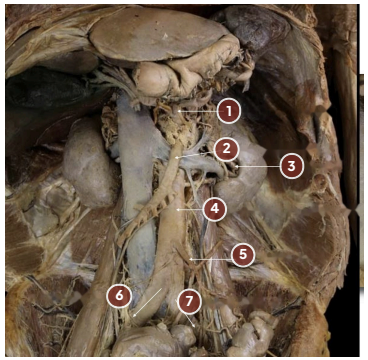
Extra notes and drawings

Blood Flow of the Abdomen and Lower Limb

Competencies 126 and 127: abdominal aorta and branches, the inferior vena cava, the hepatic portal system, pelvic and lower limb vessels.

Name _____

A. Abdominal aorta and its branches (cadaver and model) COMPETENCY 126



Cadaver, abdominal aorta and branches

1. Celi
2. Sup mes arte
3. Left arte
4. Abd aort
5. Infe mes artery
6. Right common iliac artery
7. Left common iliac artery

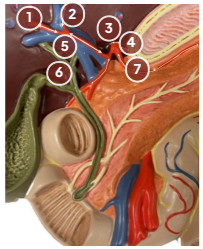
FILL IN

Unpaired, top to bottom: celiac, superior mesenteric, inferior mesenteric

Celiac trunk splits into left gastric, splenic, common hepatic

Paired branches: renal, gonadal

Ends at L4 as the common iliac arteries

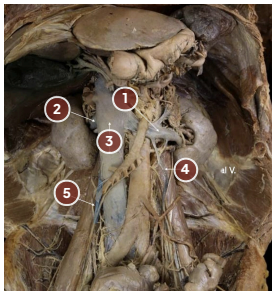


Model, celiac trunk and its branches

1. Right branch of hepatic artery proper
2. Left branch of hepatic artery proper
3. Celiac trunk
4. Left gastric artery
5. Common hepatic artery
6. Gastroduodenal artery
7. Splenic artery

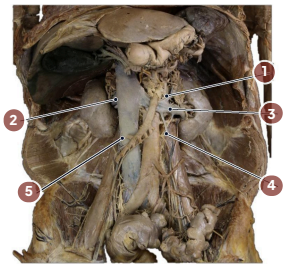
Extra notes and drawings

B. Inferior vena cava and its tributaries (cadaver) COMPETENCY 126



Cadaver, abdominal veins

1. Left renal vein
2. Right renal vein
3. Inferior vena cava
4. Left gonadal vein
5. Right gonadal vein



Cadaver, inferior vena cava and tributaries

1. Left suprarenal vein
2. Right renal vein
3. Left renal vein
4. Left gonadal vein
5. Right gonadal vein

FILL IN

IVC forms at L5 from the: common iliac veins

Through the diaphragm at: T8

Right gonadal vein drains to: the inferior vena cava

Left gonadal vein drains to: the left renal vein

Hepatic veins enter: just below the diaphragm

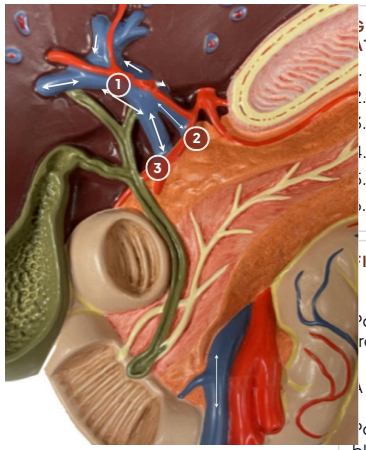
Extra notes and drawings

Blood Flow of the Abdomen and Lower Limb

Competencies 126 and 127: abdominal aorta and branches, the inferior vena cava, the hepatic portal system, pelvic and lower limb vessels.

Name _____

C. The hepatic portal system: the detour through the liver (model) COMPETENCY 126



Model, hepatic portal system

1. Hepatic portal vein
2. Splenic vein
3. Superior mesenteric vein

GLUCOSE, VILLUS TO RIGHT ATRIUM

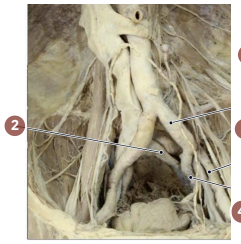
1. capillaries of a villus (bed 1)
2. superior mesenteric vein
3. hepatic portal vein
4. hepatic sinusoids (bed 2)
5. central vein, hepatic veins
6. inferior vena cava, right atrium

FILL IN

Portal vein forms from	superior mesenteric and splenic veins
A portal system has	two capillary beds in a row
Portal vein carries blood	in; hepatic veins carry it out

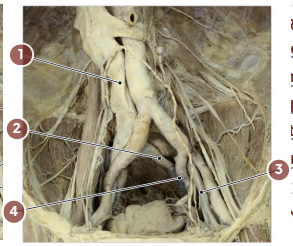
Extra notes and drawings

D. Pelvis and lower limb: one artery, renamed at each landmark COMPETENCY 127



Cadaver, iliac arteries

1. Common iliac artery
2. Common iliac vein
3. External iliac artery
4. Internal iliac artery



Cadaver, iliac veins

1. Inferior vena cava
2. Common iliac vein
3. External iliac vein
4. Internal iliac vein

RY TO THE FOOT, IN ORDER

1. ernal iliac
2. noral, below the inguinal ligament
3. pliteal, after the adductor hiatus
4. terior and posterior tibial
5. rsalis pedis; plantar arteries

N

Femoral triangle, lateral to medial: nerve, artery, vein (NAVEL)

Superficial vein up the medial leg: great saphenous

Sciatic nerve splits into tibial and common fibular

Extra notes and drawings